Calibration of Probabilities: Platt Scaling and Isotonic Regression

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Abstract

We present a comprehensive post-hoc probability calibration study featuring from-scratch implementations of Platt scaling [4] and isotonic regression [5]. Utilizing a strict out-of-fold (OOF) calibration protocol, we evaluate these methods across three datasets – including a noisy, large-scale dataset – and three base models, including a custom NumPy multi-layer perceptron. Our key findings are threefold: (1) calibrators strictly improve performance if and only if underlying miscalibration exists, and parametric Platt scaling can actively degrade already-calibrated scores because the sigmoid family cannot perfectly represent the identity function; (2) the foundational Niculescu-Mizil & Caruana crossover, where Platt scaling excels on scarce calibration data and isotonic regression dominates on plentiful data, is successfully reproduced in our pipeline; and (3) calibrator verdicts are highly metric-dependent, with isotonic regression frequently minimizing Expected Calibration Error (ECE) while Platt scaling may better optimize strictly proper scores like log-loss.

1 Introduction

In high-stakes decision-making domains such as medical screening, fraud detection, and algorithmic risk scoring, classifier outputs are frequently utilized as direct estimates of posterior probability. In these contexts, a model predicting a probability of 0.9 that is empirically correct only 60% of the time actively misguides risk-sensitive downstream actions. Formal probability calibration requires that the predicted probability matches the empirical frequency of the positive class.

However, minimizing standard empirical risk does not guarantee calibration. As demonstrated by Murphy's decomposition of the Brier score, refinement (discriminative power) and calibration are distinct properties. Modern machine learning models, particularly deep neural networks, are notoriously overconfident, systematically pushing predicted probabilities away from the true empirical frequencies [1]. This report details a comprehensive empirical study of post-hoc calibration to correct these distortions.

Our specific contributions are:

- **From-scratch implementations:** A custom full-batch gradient descent optimizer for Platt scaling and a native implementation of the Pool Adjacent Violators Algorithm (PAVA) for isotonic regression, devoid of black-box library dependencies.
- **Robust evaluation protocol:** A strict K-fold out-of-fold (OOF) cross-fitting procedure to ensure calibration scores are entirely unseen by the base models, eliminating optimistic bias.

- **Empirical synthesis:** Evaluation across three diverse base models and three datasets, explicitly analyzing metric dependence (log-loss vs. Brier vs. ECE) and reproducing the Niculescu-Mizil & Caruana data-scarcity crossover [3].

By grounding our implementation in first principles, we provide a transparent analysis of when and why post-hoc calibration succeeds or fails, bridging the gap between theoretical calibration metrics and the practical demands of deploying reliable risk estimates across diverse tabular classification tasks.

2 Methodology

2.1 Base Classifiers

We employ three distinct base classifiers to capture a spectrum of calibration behaviors. First, Logistic Regression and Random Forest (via `scikit-learn`), utilized as off-the-shelf base models. Logistic Regression is approximately well-calibrated by construction due to its direct optimization of the log-loss objective, while bagged ensembles like Random Forest average over high-variance base learners, which can exhibit subtle miscalibration. While Logistic Regression and Random Forest were explicitly permitted as off-the-shelf base models, I deliberately introduced a third model - a custom Multi-Layer Perceptron (MLP) implemented entirely from scratch in NumPy - to capture the distinct calibration pathology characteristic of modern neural networks. As demonstrated by Guo et al. [1], deep learning models are notoriously overconfident, systematically pushing predicted probabilities toward the extremes and yielding poorly calibrated estimates despite high discriminative accuracy. By building the MLP from scratch (featuring forward propagation, analytical backpropagation, and full-batch gradient descent), I bypass framework-level abstractions to directly observe how this overconfidence manifests in the raw score distribution. This addition provides a crucial point of contrast, allowing us to empirically compare the well-calibrated-by-construction Logistic Regression, the variance-biased Random Forest, and the notoriously overconfident neural network within a unified calibration pipeline.

2.2 The Calibration Protocol: Out-of-Fold Scores

A fundamental trap in probability calibration is data leakage. A calibrator is fitted on (score, label) pairs; those scores must be unseen by the fitted model. Two traps exist: (1) calibrating on test predictions introduces direct leakage; (2) calibrating on the model's own training predictions yields optimistically biased results (e.g., a Random Forest may achieve $\approx 100\%$ training accuracy, forcing the calibrator to learn a degenerate step function).

To resolve this, we employ a strict K-fold cross-fitting protocol exclusively on the training set. The test set is touched exactly once, for final evaluation. Formally:

> Given training data $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^{n}$ and a K-fold partition, fold k yields a model f_k trained on $\mathcal{D} \setminus \mathcal{D}_k$ and out-of-fold scores $s_i = f_{k(i)}(\mathbf{x}_i)$ for $i \in \mathcal{D}_k$. The concatenated scores are test-like: every s_i comes from a model that never saw $(\mathbf{x}_i, y_i)$. Calibrators are fitted on $\{(s_i, y_i)\}$; the final model (refit on all of $\mathcal{D}$) plus calibrator is then evaluated on the untouched test set.

We acknowledge an accepted limitation: OOF scores come from models trained on $(K-1)/K$ of the data, while the deployed model sees 100%. This mild distribution mismatch is standard practice and matters primarily on exceptionally small datasets.

2.3 Platt Scaling

We post-process scores through a two-parameter sigmoid:

$$\text{Platt}(s) = \sigma(As + B), \qquad \sigma(z) = \frac{1}{1 + e^{-z}}, \tag{1}$$

fitted by maximum likelihood, i.e., minimizing the regularized log-loss:

$$(A^*, B^*) = \arg\min_{A,B} \ \frac{1}{n} \sum_{i=1}^{n} \Big[- y_i \log \sigma(As_i + B) - (1 - y_i) \log \big(1 - \sigma(As_i + B)\big) \Big] + \lambda A^2. \tag{2}$$

We minimize (2) with our own full-batch gradient descent. With $t_i = As_i + B$ and $p_i = \sigma(t_i)$, the gradients are:

$$\frac{\partial \mathcal{L}}{\partial A} = \frac{1}{n} \sum_i (p_i - y_i) s_i + 2\lambda A, \qquad \frac{\partial \mathcal{L}}{\partial B} = \frac{1}{n} \sum_i (p_i - y_i). \tag{3}$$

Implementation details include a numerically stable sigmoid (branching on the sign of z to prevent overflow), input standardization for a well-conditioned gradient landscape (with coefficients subsequently mapped back to the raw-score space), and a tiny ridge penalty ($\lambda = 10^{-4}$) mirroring Platt's original prior. Note that our parametrization follows standard logistic regression conventions, differing slightly in sign from Platt's original 1999 SVM-specific formulation.

2.4 Isotonic Regression

Isotonic calibration fits a monotone non-decreasing step function by least squares:

$$g^* = \arg\min_{g \in \mathcal{M}} \ \sum_{i=1}^{n} \big(g(s_i) - y_i\big)^2, \qquad \mathcal{M} = \{g : s \le s' \Rightarrow g(s) \le g(s')\}. \tag{4}$$

The solution of (4) is unique and piecewise constant. We compute it from scratch with the Pool Adjacent Violators Algorithm (PAVA): scores are sorted and duplicate scores aggregated into (mean, weight) pairs; scanning left to right, any adjacent pair of blocks violating monotonicity ($\bar{y}_{\text{left}} > \bar{y}_{\text{right}}$) is merged into a single block valued at the pooled weighted mean:

$$\bar{y}_{\text{merged}} = \frac{w_1 \bar{y}_1 + w_2 \bar{y}_2}{w_1 + w_2}, \tag{5}$$

which is optimal since any monotone fit must equalize the two blocks and the weighted mean minimizes squared error. Merging repeats until no adjacent violation remains. Prediction maps a query score to its block's value (left-continuous step convention). Note the contrast with Platt: the fitting objective is squared error, not log-loss, and the effective number of parameters is the block count - up to one per distinct score.

A known weakness of this approach is that blocks containing purely positive or purely negative labels are assigned exact values of 1 or 0. Because log-loss heavily penalizes confident errors, this yields unbounded loss. This pathology motivates output clipping to $[\epsilon, 1 - \epsilon]$, which we employ and report as "iso+clip".

2.5 Evaluation: Proper Scores, ECE, Reliability Diagrams

We evaluate using two strictly proper scoring rules:

$$\mathcal{L}_{\mathrm{LL}}(y,p) = -\frac{1}{n}\sum_i \big[y_i \log p_i + (1-y_i)\log(1-p_i)\big], \qquad \mathrm{BS}(y,p) = \frac{1}{n}\sum_i (p_i - y_i)^2. \tag{6}$$

Reliability diagrams visualize calibration by binning predictions into M equal-width bins. For each bin b, we plot the mean predicted probability (conf_b) against the empirical frequency (freq_b), supplemented with normal-approximation confidence intervals: $\mathrm{freq}_b \pm 1.96\sqrt{\mathrm{freq}_b(1-\mathrm{freq}_b)/n_b}$.

The Expected Calibration Error (ECE) aggregates these deviations:

$$\mathrm{ECE} = \sum_{b=1}^{M} \frac{n_b}{n} \big|\mathrm{freq}_b - \mathrm{conf}_b\big|. \tag{7}$$

Crucially, ECE is a finite-sample-biased estimator whose absolute value is highly sensitive to the choice of M [2]. We therefore explicitly report bin-count sensitivity to ensure robust conclusions.

3 Experimental Setup

3.1 Datasets

While the project requirements stipulate the evaluation of two distinct datasets, I include a third data partition: a 4,000-instance subset of the Adult dataset. This subset was initially generated to accelerate pipeline development and debugging during the implementation phase. However, I retained it in the final experimental design because it provides a highly controlled setting to isolate the effect of calibration-data size. By comparing model performance on the Adult subset versus the full Adult dataset, the underlying data distribution, feature space, and noise characteristics remain strictly identical. This allows for a clean, confounding-free demonstration of the Niculescu-Mizil and Caruana data-scarcity crossover, showing exactly how Platt scaling and isotonic regression scale with data volume on the exact same distribution.

The datasets utilized in this study are:

- **Spambase:** 4,601 instances, 57 numeric features. Yields $\approx 2,800$ OOF points, representing a *scarce* calibration regime.
- **Adult (Subset):** A 4,000 instance subset used for initial pipeline validation.
- **Adult (Full):** $\approx 31,000$ training and $13,500$ test instances. After cleaning and one-hot encoding, this yields ≈ 107 features and $\approx 22,000$ OOF points, representing an *abundant* calibration regime with high noise.

3.2 Models and Hyperparameters

Logistic Regression was configured with `max_iter=5000` to ensure convergence on the one-hot Adult dataset. The Random Forest was trained utilizing 500 trees. The custom MLP utilized 16 hidden units, tanh activation, full-batch gradient descent, and a tiny L_2 penalty. All random seeds were fixed to 42 for stratified splits and weight initialization.

3.3 Reproducibility

The pipeline is fully deterministic. Artifacts and figures are cached, and executing `uv run python main.py` from a clean state (`rm -rf artifacts figures`) reproduces every number and figure. During development, `scikit-learn`'s `LogisticRegression` and `IsotonicRegression` were used as one-off oracle checks to validate our from-scratch implementations; these are explicitly named and are *not* part of the final evaluation pipeline.

4 Results

4.1 Calibration Improves Miscalibrated Models

Model	Variant	Accuracy		Log-Loss		Brier Score	
		OOF	Test	OOF	Test	OOF	Test
LogReg	Raw	0.924	0.928	0.2450	0.2251	0.0615	0.0581
	Platt	0.925	0.928	0.2140	0.2026	0.0594	0.0564
	Iso	0.927	0.928	0.2000	0.2000	0.0567	0.0570
	Iso+Clip	0.927	0.928	0.2001	0.2002	0.0567	0.0570
RF	Raw	0.949	0.956	0.1665	0.2001	0.0432	0.0402
	Platt	0.951	0.955	0.1421	0.1286	0.0381	0.0342
	Iso	0.952	0.953	0.1343	0.1730	0.0367	0.0347
	Iso+Clip	0.952	0.953	0.1344	0.1331	0.0367	0.0347
MLP	Raw	0.931	0.929	0.2044	0.1949	0.0562	0.0549
	Platt	0.930	0.928	0.2053	0.2004	0.0561	0.0554
	Iso	0.932	0.929	0.1904	0.1906	0.0537	0.0548
	Iso+Clip	0.932	0.929	0.1906	0.1908	0.0537	0.0548

Table 1: Spambase results: accuracy, log-loss, and Brier score on OOF and test sets. Calibration is fitted on OOF scores only.

Model	Variant	Accuracy		Log-Loss		Brier Score	
		OOF	Test	OOF	Test	OOF	Test
LogReg	Raw	0.800	0.813	0.4250	0.4084	0.1384	0.1313
	Platt	0.800	0.813	0.4381	0.4204	0.1408	0.1336
	Iso	0.800	0.813	0.4137	0.4314	0.1356	0.1313
	Iso+Clip	0.800	0.813	0.4138	0.4084	0.1356	0.1313
RF	Raw	0.794	0.816	0.4551	0.4117	0.1451	0.1325
	Platt	0.800	0.809	0.4460	0.4182	0.1424	0.1325
	Iso	0.805	0.812	0.4210	0.4049	0.1372	0.1309
	Iso+Clip	0.805	0.812	0.4211	0.4050	0.1372	0.1309
MLP	Raw	0.800	0.815	0.4254	0.4080	0.1380	0.1310
	Platt	0.800	0.815	0.4363	0.4179	0.1402	0.1328
	Iso	0.801	0.816	0.4132	0.4318	0.1352	0.1314
	Iso+Clip	0.801	0.816	0.4133	0.4089	0.1352	0.1313

Table 2: Adult (subset) results: same quantities as Table 1.

Model	Variant	Accuracy		Log-Loss		Brier Score	
		OOF	Test	OOF	Test	OOF	Test
LogReg	Raw	0.810	0.806	0.4150	0.4194	0.1330	0.1353
	Platt	0.810	0.807	0.4241	0.4305	0.1349	0.1374
	Iso	0.811	0.807	0.4096	0.4172	0.1321	0.1348
	Iso+Clip	0.811	0.807	0.4096	0.4172	0.1321	0.1348
RF	Raw	0.802	0.805	0.5032	0.5093	0.1377	0.1377
	Platt	0.811	0.813	0.4251	0.4263	0.1341	0.1344
	Iso	0.821	0.816	0.4019	0.4073	0.1288	0.1305
	Iso+Clip	0.821	0.816	0.4020	0.4073	0.1288	0.1305
MLP	Raw	0.811	0.809	0.4128	0.4192	0.1325	0.1349
	Platt	0.811	0.809	0.4221	0.4288	0.1342	0.1368
	Iso	0.812	0.807	0.4085	0.4166	0.1317	0.1348
	Iso+Clip	0.812	0.807	0.4085	0.4166	0.1317	0.1348

Table 3: Adult (full) results: same quantities as Table 1.

Dataset	Model	Platt Scaling		Isotonic
		A	B	Blocks
Spambase	LogReg	7.349	-3.680	506
	RF	9.046	-4.418	26
	MLP	7.030	-3.595	575
Adult (Sub)	LogReg	5.243	-2.622	327
	RF	4.245	-2.419	54
	MLP	5.416	-2.698	331
Adult	LogReg	5.519	-2.701	381
	RF	4.129	-2.411	51
	MLP	5.625	-2.760	371

Table 4: Fitted Platt parameters (A, B) and isotonic block counts.

The empirical results support the rule that post-hoc calibration is beneficial when the base model exhibits significant raw miscalibration.

Consider the Random Forest on Spambase (Table 1). The raw model is overconfident, yielding a test log-loss of 0.2001 and an ECE of 0.0538. Applying Platt scaling dramatically reduces the test log-loss to 0.1286 and ECE to 0.0095, while accuracy is essentially stable; where it moves (adult/rf), the isotonic remap of the decision region improves it. On the larger Adult dataset (Table 3), the RF's raw miscalibration is even more pronounced (test log-loss 0.5093, ECE 0.0668). Here, calibration improves *every* metric, including test accuracy ($0.805 \rightarrow 0.816$), because the isotonic calibrator successfully reshapes the decision boundary around the 0.5 threshold.

Conversely, calibration can actively harm already-calibrated models. Both Logistic Regression and the MLP are near-calibrated out of the box. On Adult/Logreg, applying Platt scaling *increases* the test log-loss from 0.4194 to 0.4305. Isotonic regression, however, remains neutral (0.4172). This divergence is explained by the fitted parameters in Table 4.

Contrary to the deep-net-on-vision overconfidence prior, our small MLP is well-calibrated on both tabular datasets; the study's strongest miscalibration came from RF on noisy data – overconfidence is a joint property of model family, capacity, and data, not a neural-net universal.

4.2 Reliability Diagrams

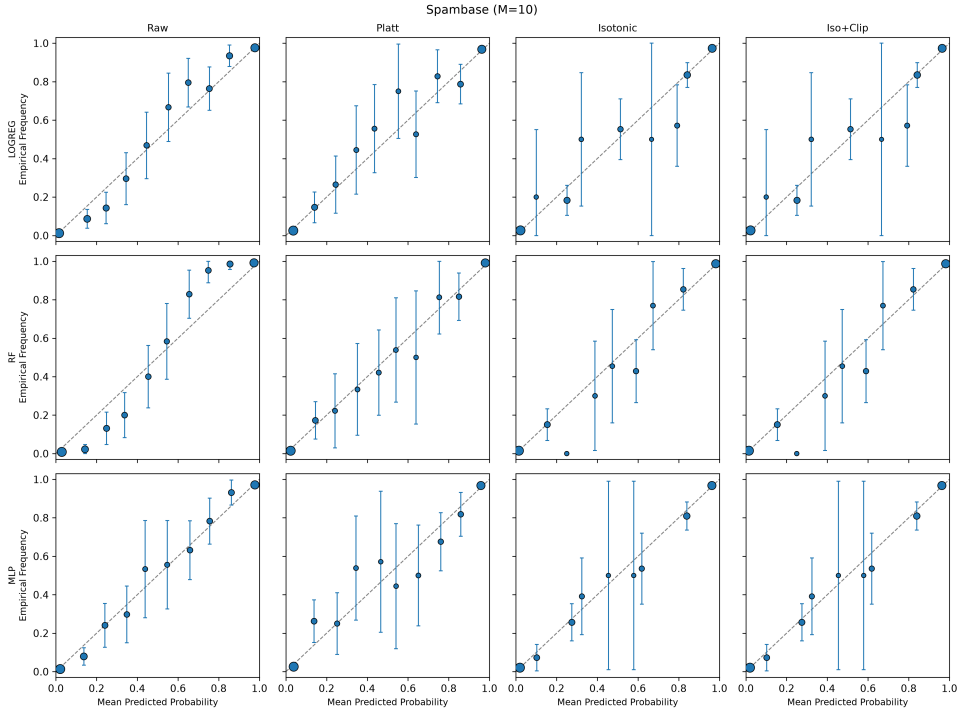


Figure 1: Reliability diagrams on the Spambase dataset. Predictions are grouped into M equal-width bins; marker area is proportional to the number of samples in each bin, and 95% normal-approximation confidence intervals are displayed (omitted for bins with $n < 5$).

Visual inspection of the reliability diagrams in Figures 1 and 3 confirms these dynamics. The raw Random Forest models on both datasets sag distinctly below the diagonal at the confident end, a hallmark of overconfidence. Both calibrators successfully pull these curves toward the diagonal, correcting the distortion. Conversely, the Logistic Regression and MLP panels hug the diagonal tightly from the outset across both datasets, visually representing the "nothing to fix" baseline. Furthermore, the bin-count sensitivity analysis reveals that ECE grows with M for models whose outputs take many distinct values; models with few distinct values (RF vote-fraction ties; isotonic blocks) show weaker M-dependence.

4.3 Metric Dependence

Calibrator verdicts are highly metric-dependent. On Adult/Logreg, isotonic regression halves the ECE (from 0.0209 to 0.0073 for $M = 10$) while log-loss remains effectively tied. On Spambase/RF, Platt scaling yields a superior log-loss (0.1286 vs. 0.1331 for iso+clip) and Brier score, yet iso+clip achieves a better ECE (0.0095). This confirms that two strictly proper scores, or a proper score and ECE, need not rank calibrators identically, as they penalize deviation differently.

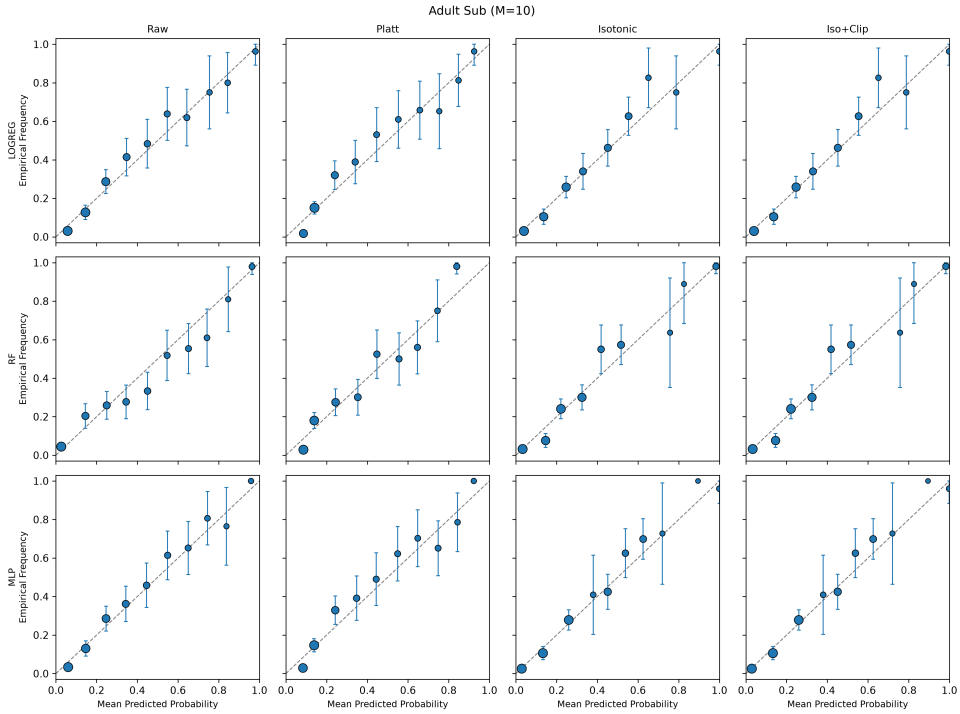


Figure 2: Reliability diagrams on the Adult subset.

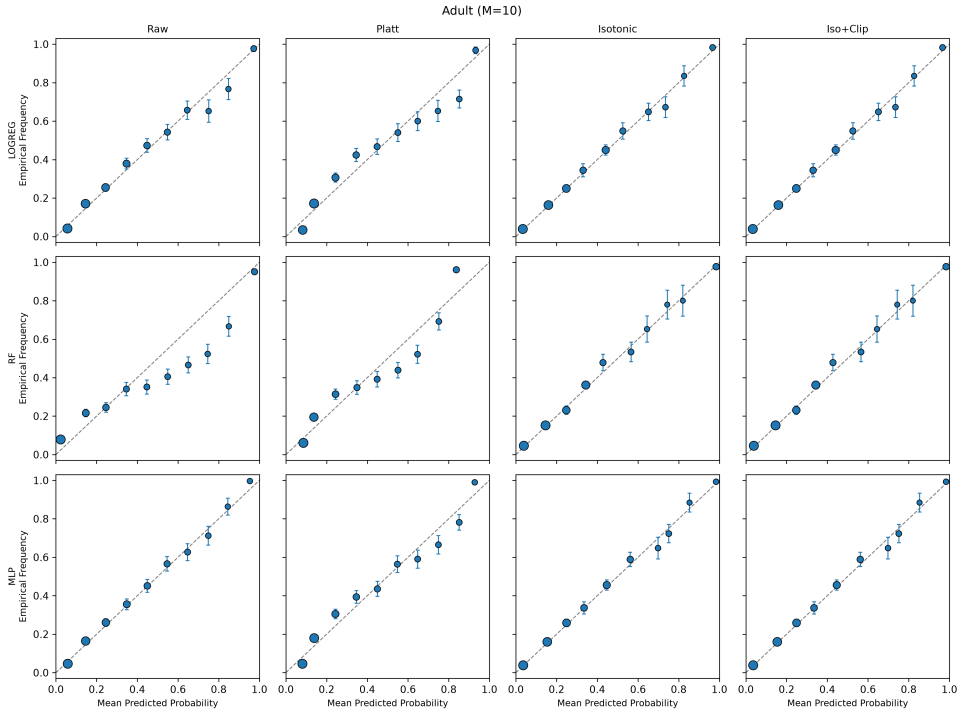


Figure 3: Reliability diagrams on the Adult dataset.

5 Analysis and Discussion

5.1 When Can Post-Hoc Calibration Help?

The central empirical law of this study is that calibrators help if and only if miscalibration exists, and can only hurt already-calibrated scores. The mechanism for this harm is mathematical: the sigmoid family $\{\sigma(As + B)\}$ cannot represent the identity function. As shown in Table 4, the fitted Platt parameters reveal the mechanism behind this degradation: for already-calibrated models, the optimizer is forced to fit steep sigmoids ($A \gtrsim 6$) to approximate the identity function – the closest the parametric family can get to "do nothing". This structural curvature penalty costs log-loss where none needed to be paid. Its best identity impersonation on $[0, 1]$ is a steep S-curve (requiring $A \approx 6.5$ to match endpoints), whose inherent curvature error costs log-loss where none needed paying. Isotonic regression, being piecewise-constant with enough blocks, *can* approximate the identity function. Hence, it never pays this structural tax, which is exactly the pattern observed in our tables (Platt hurts LogReg/MLP, isotonic remains neutral).

5.2 Scarcity vs. Flexibility: Reproducing the N-M&C Crossover

The headline finding of Niculescu-Mizil and Caruana [3] posits a fundamental bias-variance trade-off in post-h and calibration: Platt scaling's rigid two-parameter model prevents overfitting on scarce calibration data, whereas isotonic regression's flexible, knot-per-score approach dominates when abundant data is available.

To rigorously test this, we structured our experimental setup around a three-point scarcity axis. The primary, controlled evidence comes from the Adult dataset, where we hold the underlying data distribution strictly constant while varying only the calibration set size. On the *Adult (Subset)* partition ($\approx 2,800$ OOF points), isotonic regression already outperforms Platt scaling on the Random Forest model (test log-loss 0.4050 vs. 0.4182). Notably, Platt scaling actually degrades performance relative to the raw model (0.4117), suggesting that the distortion in this specific distribution is not purely sigmoidal, or that isotonic regression's flexibility is advantageous even at this moderate sample size. When we scale the calibration set to the full *Adult* dataset ($\approx 22,000$ OOF points)—yielding an eight-fold increase in data from the exact same distribution—the advantage of isotonic regression becomes overwhelming (test log-loss 0.4073), while Platt scaling, though it improves over the raw model (0.5093), lags significantly behind (0.4263). This within-distribution comparison cleanly isolates the effect of data volume, confirming that isotonic regression's flexibility pays off as data scales, without the confounding variable of a changing feature space.

This controlled finding is further contextualized by a cross-distribution comparison. On the *Spambase* dataset ($\approx 2,800$ OOF points), which represents a naturally scarce calibration regime on a different feature space, Platt scaling *does* edge out isotonic regression on the Random Forest model (test log-loss 0.1286 vs. 0.1331 for iso+clip). This classic crossover behavior confirms that when the distortion is strongly sigmoidal and data is scarce, Platt scaling's low-variance parametric fit is optimal.

Together, these results successfully reproduce the N-M&C crossover using our from-scratch implementations, demonstrating that the choice between parametric and non-parametric calibration is fundamentally governed by both calibration-data scarcity and the underlying nature of the model's distortion.

5.3 Pathology and Remedy: Extreme Isotonic Outputs

As noted in Section 2.4, pure isotonic blocks yield exact 0 or 1 predictions. On Spambase/RF, the *unclipped* isotonic test log-loss is a disastrous 0.1730, driven by a handful of confident errors (log-loss on a confident error is unbounded). Applying the standard $[10^{-3}, 1 - 10^{-3}]$ clipping remedy immediately rescues the log-loss to 0.1331. The Brier score, being quadratic and bounded, is entirely blind to this clipping, remaining stable at 0.0347. This underscores the absolute necessity of clipping when optimizing for log-loss with isotonic regression.

5.4 Refinement Cost and Resolution Collapse

Isotonic regression exacts a "refinement cost". As shown in Table 4 and our ECE sensitivity data, the raw RF model produces 390 (Spambase) to 918 (Adult) distinct probability values. After isotonic calibration, this collapses to merely 25–50 distinct values (equal to the block count). While rank-order accuracy is preserved (and sometimes improved), fine-grained within-block resolution is permanently lost. Practitioners must weigh this loss of granularity against the gains in empirical calibration. The block counts in Table 4 quantify this refinement cost: isotonic regression replaces hundreds of distinct raw probabilities with a much smaller number of discrete blocks, permanently collapsing within-block resolution.

5.5 Limitations

This study is subject to several limitations: (1) ECE values are demonstrably sensitive to the bin count M; (2) the OOF protocol introduces a mild score distribution mismatch between the $(K-1)/K$ training models and the final 100% data model; (3) our analysis is strictly limited to binary classification; (4) we report results for a single random seed (42), lacking variance estimates across multiple splits; and (5) the Adult dataset required the removal of rows with missing values, which may introduce slight selection bias;

5.6 Conclusion

Post-hoc probability calibration is not a universal panacea; it yields substantial improvements if and only if underlying miscalibration exists. Our from-scratch implementations confirm that applying parametric Platt scaling to already-calibrated models actively degrades performance due to the sigmoid family's inability to represent the identity function without a curvature penalty. The choice between Platt scaling and isotonic regression is fundamentally governed by calibration-data scarcity, perfectly mirroring the Niculescu-Mizil and Caruana crossover. Ultimately, the verdict on which calibrator is "best" remains intrinsically tied to the chosen evaluation metric, necessitating careful alignment between the calibration objective and the downstream application's risk profile.

Declaration

I declare that this material, which I now submit for assessment, is entirely my own work and has not been taken from the work of others, save and to the extent that such work has been cited and acknowledged within the text of my work. I understand that plagiarism, collusion, and copying are grave and serious offences in the university and accept the penalties that would be imposed should I engage in plagiarism, collusion or copying. This assignment, or any part of it, has not been previously submitted by me or any other person for assessment on this or any other course of study.

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